HMD12

Dual H-bridge motor driver with SPI configuration and current sensing

Features

- Motor supply VM: 4.5 V to 18 V (absolute maximum 20 V)
- Two H-bridges: 2.5 A peak and 1.8 A RMS per channel
- RDS(on) (high + low side): 220 mΩ typical at 25 °C
- PWM frequency up to 100 kHz
- SPI configuration interface, clock up to 5 MHz
- Analog current-sense output per channel
- Overcurrent trip: 3.4 A typical
- Thermal shutdown at 165 °C
- Logic supply VCC: 3.0 V to 5.5 V; standby current 8 µA
- TSSOP-20 package with exposed pad

General description

The HMD12 drives two brushed DC motors or one stepper motor. Fault conditions are reported on an open-drain FAULT pin and in the SPI status register.

Applications

Robotics, Printers, Camera gimbals, Valves and actuators.

Ordering information

Order code	Package	Packing	Temperature grade
HMD12-TSSOP20-T	TSSOP-20 (EP)	Tape & reel, 2500	–40 °C to 125 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
VM to GND	–0.3	20	V
VCC to GND	–0.3	6.0	V
Output current (peak, per channel)	—	2.5	A
Junction temperature TJ	—	150	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
Motor supply VM	4.5	12	18	V
Logic supply VCC	3.0	3.3	5.5	V
PWM frequency	—	—	100	kHz
SPI clock	—	—	5	MHz

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
Output on-resistance (HS + LS)	TJ = 25 °C, I = 1 A	—	220	300	mΩ
Overcurrent trip threshold		2.9	3.4	3.9	A
Thermal shutdown		—	165	—	°C
Standby current (VM)	Sleep mode	—	8	15	µA
Current-sense gain		—	0.5	—	V/A

Pin configuration and functions

Pin	Name	Type	Description
1	OUT1A	Output	Bridge 1, output A.
2	OUT1B	Output	Bridge 1, output B.
5	VM	Power	Motor supply, 4.5 V to 18 V.
8	VCC	Power	Logic supply.
12	SCK	Input	SPI clock.
13	SDI	Input	SPI data in.
15	FAULT	Output	Open-drain fault flag.

Package and thermal information

Package	Thermal resistance θJA	Dimensions
TSSOP-20 (EP)	θJA = 32 °C/W	6.5 × 4.4 mm

Application information

Add at least 47 µF of bulk capacitance close to VM. Solder the exposed pad to a large ground plane for rated thermal performance.

Revision history

Revision	Date	Change
Rev A	2024-07-01	Initial release